



Department of Artificial Intelligence and Data Science

1st TUT : VII SEM 2025-2026

Course Name & Code: Statistical Machine Learning For Data Science (BAD702)

By Sunil Naidu G R

Tutorial SML for 1st CIE 2025-2026

Module 1

1. Explain Exploratory Data Analysis (EDA). Why is EDA important? **5M**
2. Explain Key Terms for Data Types. **5M**
3. Explain Estimates of Location. Find the mean, median , SD, Trimmed Mean for the Data set =[10 15 20 25 30 35 40 45 50]. Write python code for Mean, median, SD. **10M**
4. What is percentile and IQR .write the python code for percentile and IQR w.r.t population as an example. **5M**
5. Define key terms in Exploring the Data Distribution. **5M**
6. What are Density Plots and Estimates. explain with figure and python code for 1,00,000(murder rate) **5M**
7. What is Correlation and Scatterplot .find the correlated value for v1: {1, 2, 3} v2: {4, 5, 6} **5M**
8. Define hexagonal binning with Python code .write python code for tax-assessed value versus finished square feet **10M**

Module -2

1. What is sampling w.r.t population? Explain Random Sampling and Sample Bias. **5M**
2. Define Key Terms for Random Sampling. **5M**
3. Explain bias w.r.t Scatterplot of shots from a gun with true aim & Scatterplot of shots from a gun with biased aim. **5M**
4. Explain Regression to the Mean. Draw a plot of Galton's study that identified the phenomenon of regression to the mean. **5M**
5. Define keyterms of Sampling Distribution of a Statistic. **5M**



Estd : 1986



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6. Draw Histogram of annual incomes of 1,000 loan applicants ,Mean 5 and Mean 20 . python code uses seaborn's FacetGrid to show the three histograms . **10M**
7. Explain Central Limit Theorem and S.E .Find S.E for Data set =[10 15 20 25 30 35 40 45 50] **10M**
8. What is Bootstrap. Write the major Python packages don't provide implementations of the bootstrap approach **5M**
9. Explain Key Terms for Binomial Distribution. **5M**